Day - 21
Reading CSV Files
in Python

SKILL COURSE

A.I. Powered 30 Days Python Micro Course By Satish Dhawale (Microsoft Certified Trainer)

What is a CSV File?

Layman Explanation

CSV = **Comma Separated Values**
It is like an Excel sheet but stored as plain text.

Each row = one record
Each column = one field
Separated by commas.

Technical Explanation

CSV is a structured text file where each line contains values separated by commas:

Order_ID,Customer,City,Product,Amount

ORD101,Rohit,Mumbai,Laptop,55000

Why CSV is essential for Data Analysts?

- 90% real-world datasets come in CSV
- Used in Excel, Tableau, Python, Power BI
- Lightweight
- Easy to import/export
- Works great with Python & Pandas

Reading CSV in Python (Without Pandas)

```
import csv

with open("day21_sales_data.csv", "r") as file:
    reader = csv.reader(file)

    for row in reader:
        print(row)
```

Skip Header Row

```
import csv

with open("day21_sales_data.csv", "r") as f:
    reader = csv.reader(f)
    next(reader)  # Skip header

    for row in reader:
        print(row)
```

Read as Dictionary (BEST PRACTICE)

```
import csv

with open("day21_sales_data.csv", "r") as f:
    reader = csv.DictReader(f)

    for row in reader:
        print(row["Customer_Name"], "-", row["City"])
```

Practical Data Analyst Tasks

```
import csv

total = 0

with open("day21_sales_data.csv") as f:
    reader = csv.DictReader(f)

    for row in reader:
        total += int(row["Price"])

print("Total Sales =", total)
```

Filter all Sales from Mumbai

```
sales_mumbai = []

with open("day21_sales_data.csv") as f:
    reader = csv.DictReader(f)

    for row in reader:
        if row["City"] == "Mumbai":
            sales_mumbai.append(row)

print(sales_mumbai)
```

Count Orders Per City

```
count = {}

with open("day21_sales_data.csv") as f:
    reader = csv.DictReader(f)

    for row in reader:
        city = row["City"]
        count[city] = count.get(city, 0) + 1

print(count)
```